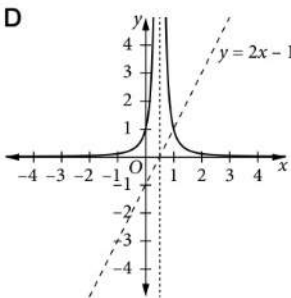
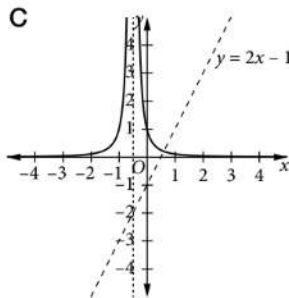
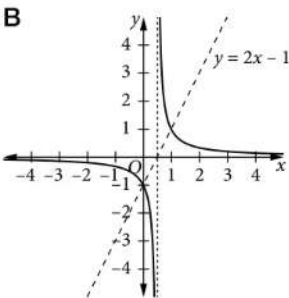
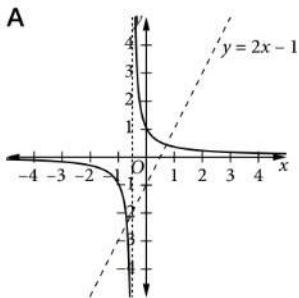
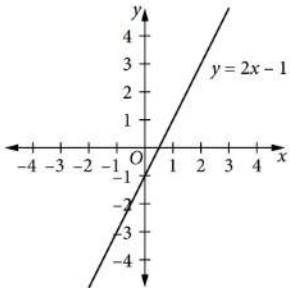


EXERCISE 17.1 RECIPROCAL FUNCTIONS

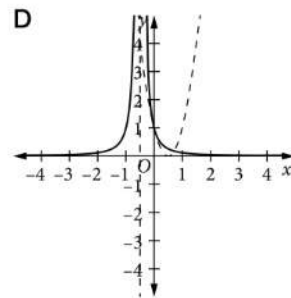
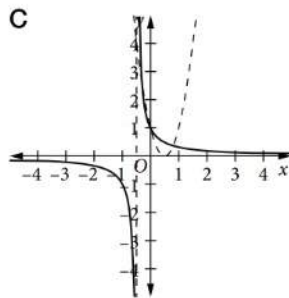
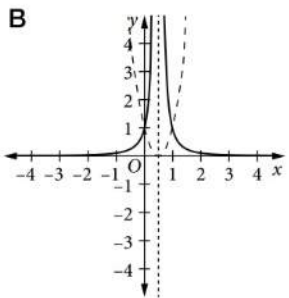
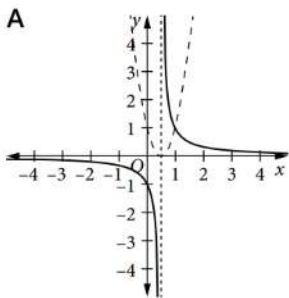
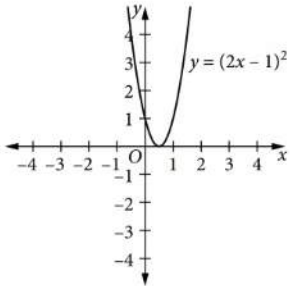
1 The graph of $y = 2x - 1$ is shown.

Which of the following represents the graph of $y = \frac{1}{2x-1}$?



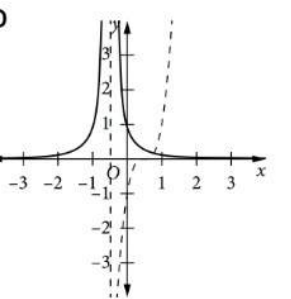
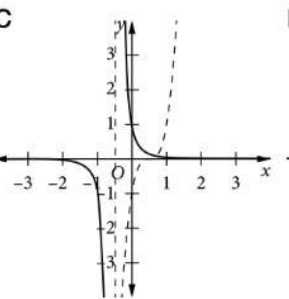
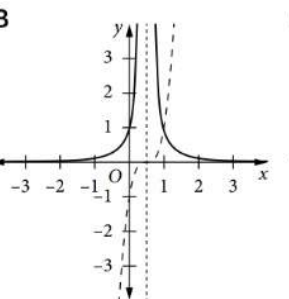
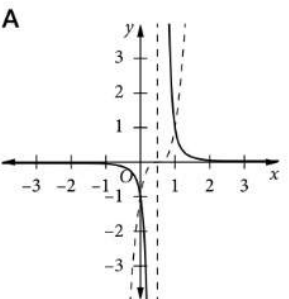
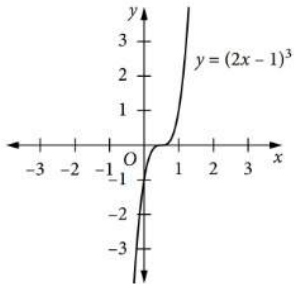
2 The graph of $y = (2x - 1)^2$ is shown.

Which of the following represents the graph of $y = \frac{1}{(2x-1)^2}$?

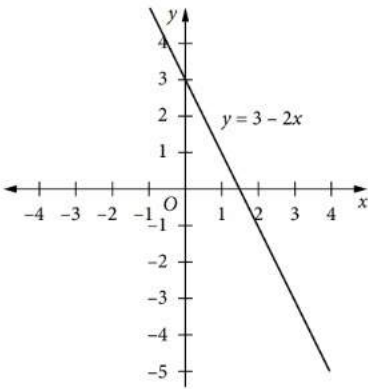


3 The graph of $y = (2x - 1)^3$ is shown.

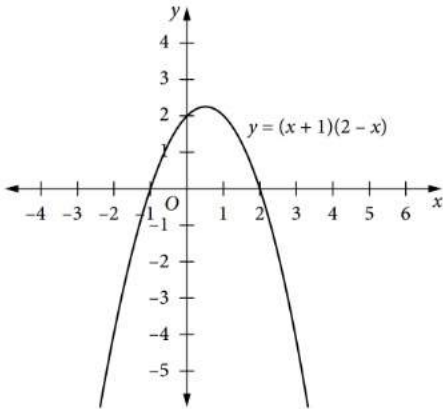
Which of the following represents the graph of $y = \frac{1}{(2x-1)^3}$?



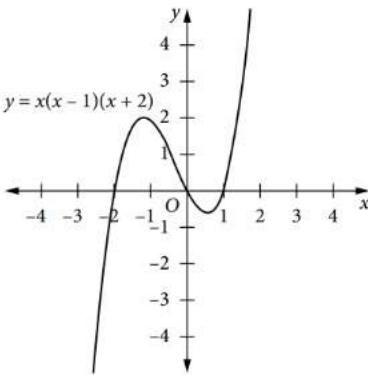
4 Given the graph of $y = 3 - 2x$, draw the graph of $y = \frac{1}{3 - 2x}$.



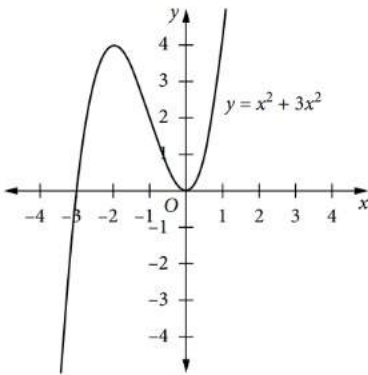
5 Given the graph of $y = (x + 1)(2 - x)$, draw the graph of $y = \frac{1}{(x + 1)(2 - x)}$.



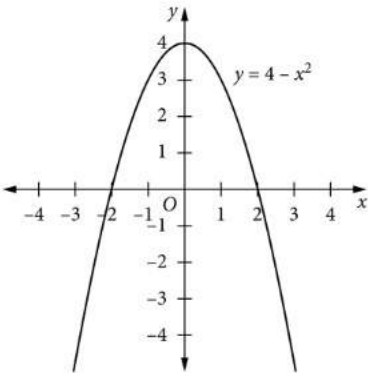
6 Given the graph of $y = x(x - 1)(x + 2)$, draw the graph of $y = \frac{1}{x(x - 1)(x + 2)}$.



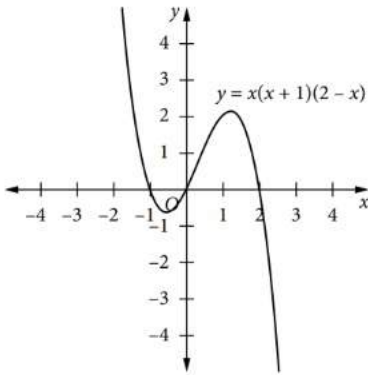
7 Given the graph of $y = x^3 + 3x^2$, draw the graph of $y = \frac{1}{x^3 + 3x^2}$.



8 Given the graph of $y = 4 - x^2$, draw the graph of $y = \frac{1}{4 - x^2}$.

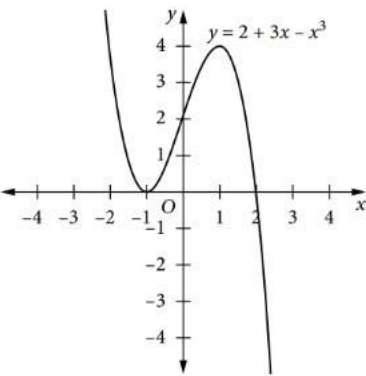


9 Given the graph of $y = x(x + 1)(2 - x)$, draw the graph of $y = \frac{1}{x(x + 1)(2 - x)}$.



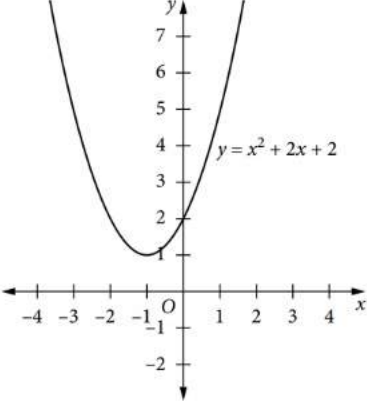
10 Given the graph of $y = 2 + 3x - x^3$, draw the graph

of $y = \frac{1}{2 + 3x - x^3}$.



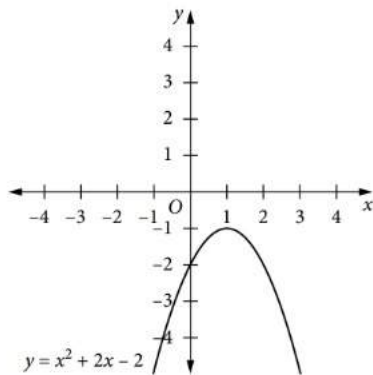
11 Given the graph of $y = x^2 + 2x + 2$, draw the graph

of $y = \frac{1}{x^2 + 2x + 2}$.



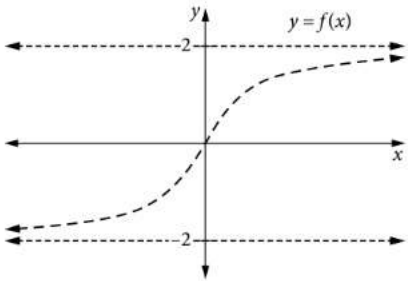
12 Given the graph of $y = -x^2 + 2x - 2$, draw the graph

of $y = \frac{1}{-x^2 + 2x - 2}$.

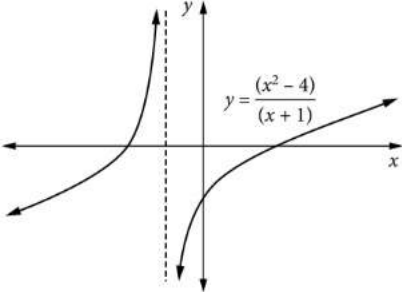


13 Given the graph of $y = f(x)$, draw the graph of

$y = \frac{1}{f(x)}$, showing the key features clearly.

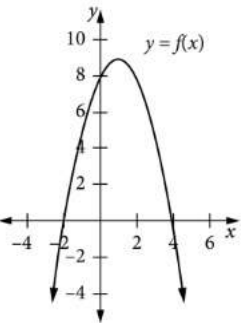


14 Given the graph of $y = \frac{x^2 - 4}{x + 1}$, draw the graph of the reciprocal function.



15 Given the graph of $y = f(x)$, draw the graph of

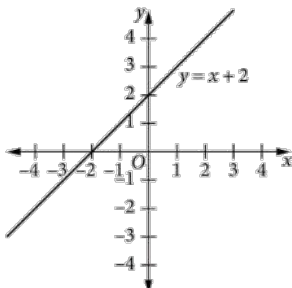
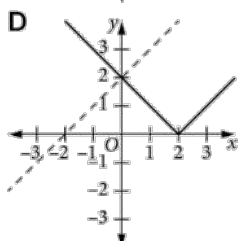
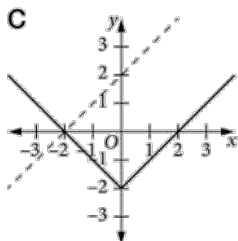
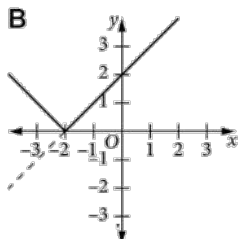
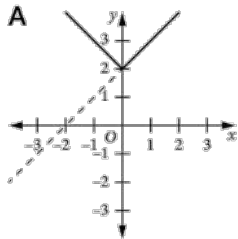
$y = \frac{1}{f(x)}$, showing the key features clearly.



EXERCISE 17.2 ABSOLUTE VALUE FUNCTIONS

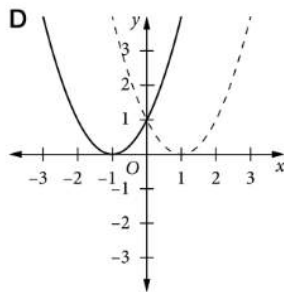
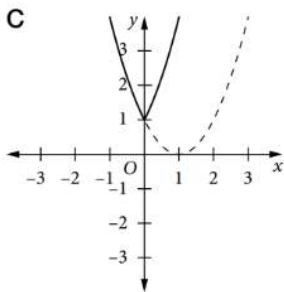
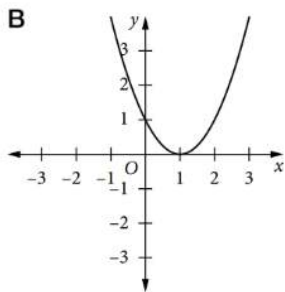
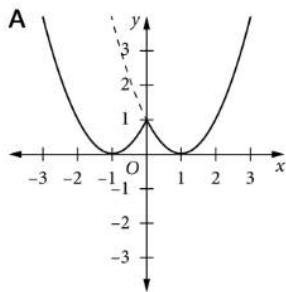
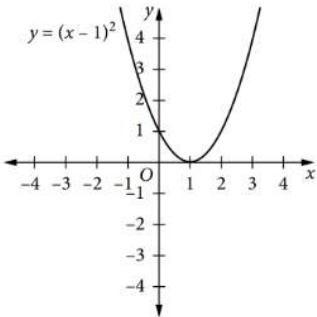
1 The graph of $y = x + 2$ is shown.

Which of the following represents the graph of $y = |x + 2|$?



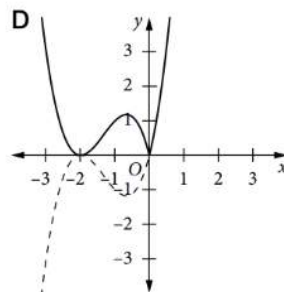
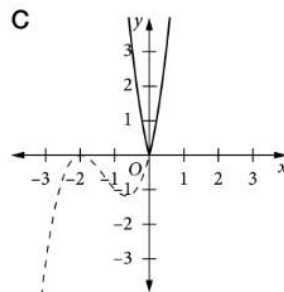
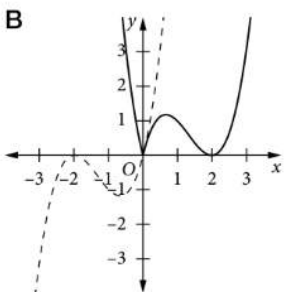
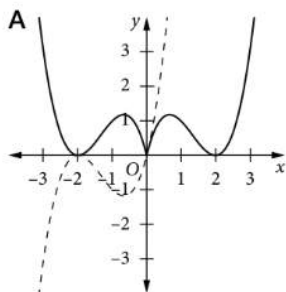
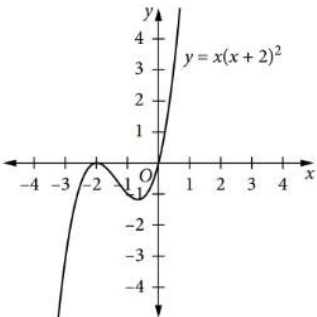
2 The graph of $y = (x - 1)^2$ is shown.

Which of the following represents the graph of $y = (|x| - 1)^2$?



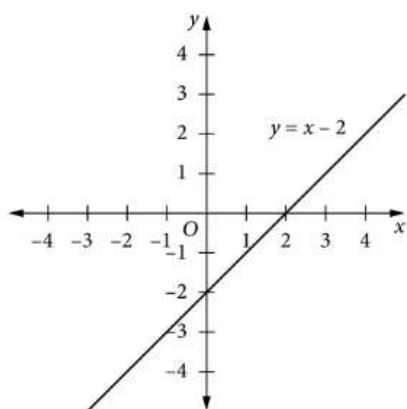
3 The graph of $y = x(x + 2)^2$ is shown.

Which of the following represents the graph of $y = |x|(|x| + 2)^2$?



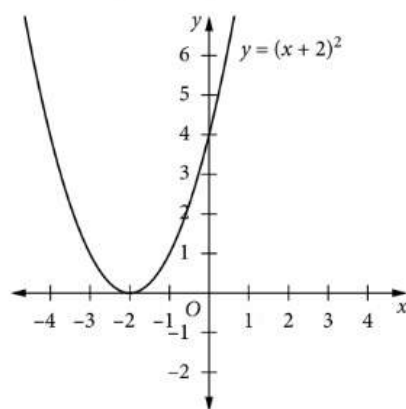
4 Given the graph of $y = x - 2$, draw:

(a) $y = |x - 2|$ (b) $y = |x| - 2$



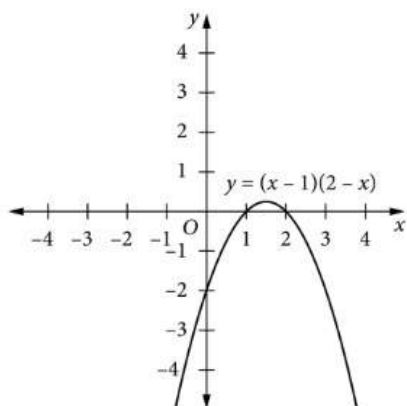
5 Given the graph of $y = (x + 2)^2$, draw:

(a) $y = |(x + 2)^2|$ (b) $y = (|x| + 2)^2$



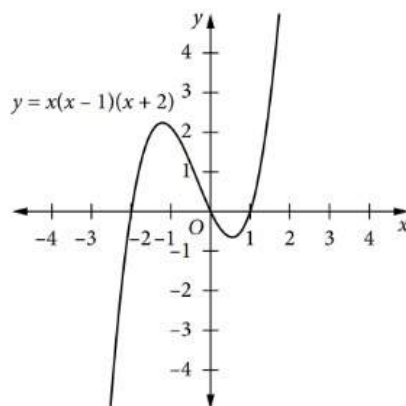
6 Given the graph of $y = (x - 1)(2 - x)$, draw:

(a) $y = |(x - 1)(2 - x)|$
(b) $y = (|x| - 1)(2 - |x|)$



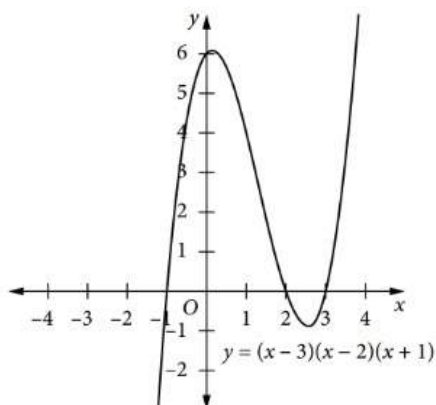
7 Given the graph of $y = x(x - 1)(x + 2)$, draw:

(a) $y = |x(x - 1)(x + 2)|$
(b) $y = |x|(|x| - 1)(|x| + 2)$



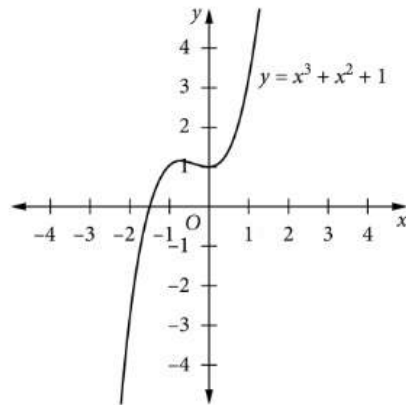
8 Given the graph of $y = (x - 3)(x - 2)(x + 1)$, draw:

(a) $y = |(x - 3)(x - 2)(x + 1)|$
(b) $y = (|x| - 3)(|x| - 2)(|x| + 1)$



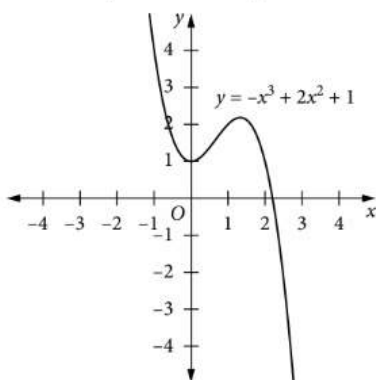
9 Given the graph of $y = x^3 + x^2 + 1$, draw:

(a) $y = |x^3 + x^2 + 1|$
(b) $y = |x|^3 + |x|^2 + 1$



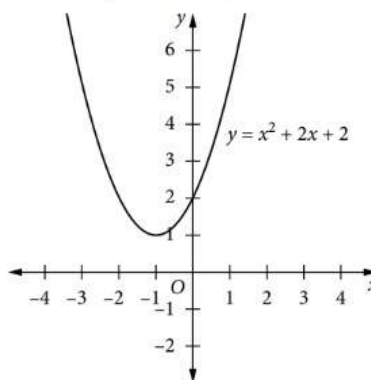
- 10** Given the graph of $y = -x^3 + 2x^2 + 1$, draw:

(a) $y = |-x^3 + 2x^2 + 1|$ (b) $y = -|x|^3 + 2|x|^2 + 1$



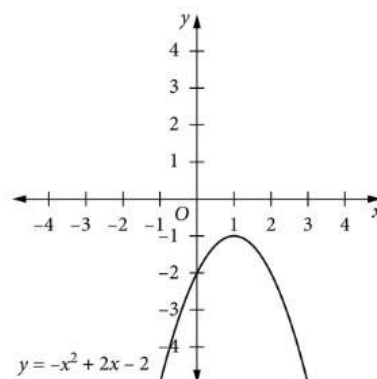
- 11** Given the graph of $y = x^2 + 2x + 2$, draw:

(a) $y = |x^2 + 2x + 2|$ (b) $y = |x|^2 + 2|x| + 2$



- 12** Given the graph of $y = -x^2 + 2x - 2$, draw:

(a) $y = |-x^2 + 2x - 2|$ (b) $y = -|x|^2 + 2|x| - 2$



- 13** Consider the function $f(x) = x^2 - c^2$, where c is a positive real number.

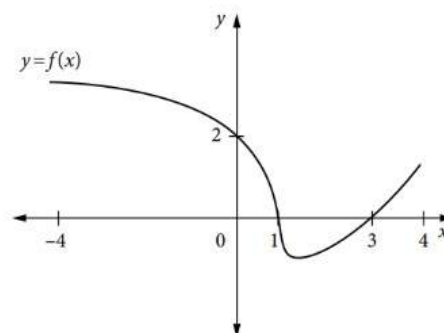
Sketch the graph of $y = \frac{1}{|f(x)|}$, showing all important features including the turning point(s), intercept(s) and asymptote(s).

- 14** The diagram shows the graph of $y = f(x)$ for $[-4, 4]$.

The graph has x -intercepts at $x = 1$ and $x = 3$, and a y -intercept at $y = 2$.

(a) Sketch the graph of $y = f(|x|)$.

(b) Sketch the graph of $y = \frac{1}{f(x)}$.

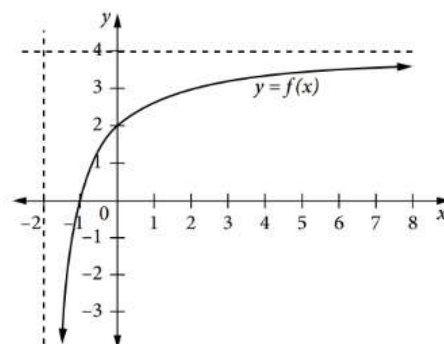


- 15** The diagram shows the graph of the curve $y = f(x)$

where $f(x) = \frac{4x+4}{x+2}$, $x > -2$.

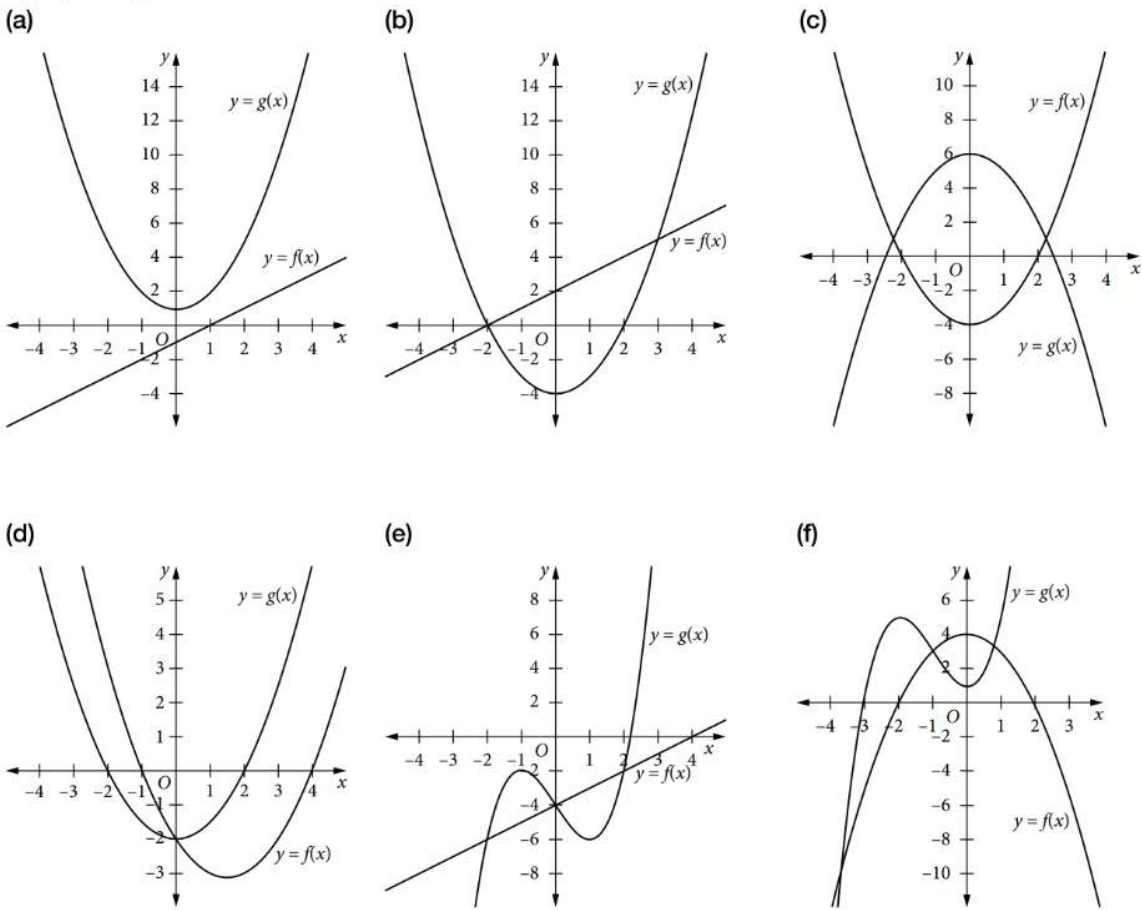
Sketch the graphs of the following curves, showing clearly the intercepts on the axes and the equations of any asymptotes.

(a) $y = |f(x)|$ (b) $y = f(|x|)$

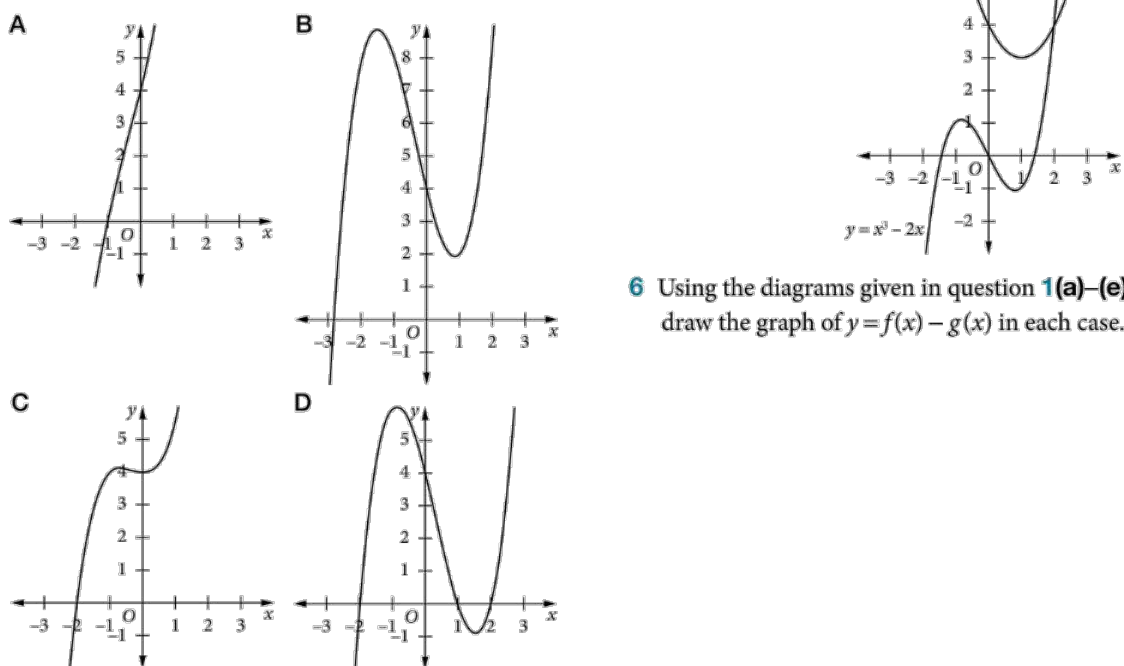


EXERCISE 17.3 GRAPHING POLYNOMIALS BY ADDING OR SUBTRACTING ORDINATES

1 The graphs of $y = f(x)$ and $y = g(x)$ are shown. By drawing vertical lines and adding ordinates, draw the graph of $y = f(x) + g(x)$. Comment on the new curve.



- 2 On the same diagram, sketch the graphs of $y = x^2 + 2x$ and $y = 1 - 2x$. Use these graphs to sketch $y = x^2 + 1$.
- 3 On the same diagram, sketch the graphs of $y = x^2 - 2x$ and $y = x^3$. Use these graphs to sketch $y = x^3 + x^2 - 2x$.
- 4 On the same diagram, sketch the graphs of $y = x^2 - 3x$ and $y = x^3 - 3x$. Use these graphs to sketch $y = x^3 + x^2 - 6x$.
- 5 The graphs of $y = x^2 - 2x + 4$ and $y = x^3 - 2x$ are shown. Which diagram represents the graph of $y = x^3 + x^2 - 4x + 4$?



6 Using the diagrams given in question 1(a)–(e), draw the graph of $y = f(x) - g(x)$ in each case.

EXERCISE 17.4 GRAPHING RECIPROCAL TRIGONOMETRIC FUNCTIONS

- 1 Using a graph of $y = \sin \theta$:
 - (a) Draw the graph of $y = \operatorname{cosec} \theta$ for $0^\circ \leq \theta \leq 360^\circ$.
 - (b) Draw the graph of $\operatorname{cosec} \theta$ for $0^\circ \leq \theta \leq 360^\circ$ using graphing software and compare it with your answer to (a).
- 2 Using a graph of $y = \cos \theta$:
 - (a) Draw the graph of $\sec \theta$ for $0^\circ \leq \theta \leq 360^\circ$.
 - (b) Draw the graph of $\sec \theta$ for $0^\circ \leq \theta \leq 360^\circ$ using graphing software and compare it with your answer to (a).
- 3 Using a graph of $y = \tan \theta$:
 - (a) Draw the graph of $\cot \theta$ for $0^\circ \leq \theta \leq 360^\circ$.
 - (b) Draw the graph of $\cot \theta$ for $0^\circ \leq \theta \leq 360^\circ$ using graphing software and compare it with your answer to (a).
- 4 Repeat questions 1 to 3 using a domain of $-180^\circ \leq \theta \leq 180^\circ$.
- 5 Sketch the graph of $y = \sin x$, $-\pi \leq x \leq \pi$. On the same axes, sketch the graph of:
 - (a) $y = -\sin x$
 - (b) $y = 1 - \sin x$
 - (c) $y = -\cos x$
- 6 Sketch the graph of each of the following, stating the period and amplitude of the function:
 - (a) $y = \sin \pi x$, $-1 \leq x \leq 1$
 - (b) $y = \cos 2\pi x$, $0 \leq x \leq 2$
 - (c) $y = \tan \frac{\pi x}{2}$, $0 \leq x \leq 2$
- 7 Sketch the graph of $y = \tan x$, $-\pi \leq x \leq \pi$. On the same axes, sketch the graph of:
 - (a) $y = 2 \tan x$
 - (b) $y = -2 \tan x$
 - (c) $y = 2 - 2 \tan x$
- 8 Sketch the graph of $y = \operatorname{cosec} x$, $-\pi \leq x \leq \pi$. On the same axes, sketch the graph of:
 - (a) $y = -\operatorname{cosec} x$
 - (b) $y = 1 - \operatorname{cosec} x$
 - (c) $y = 1 + \operatorname{cosec} x$
- 9 Sketch the graph of $y = \sec x$, $-\pi \leq x \leq \pi$. On the same axes, sketch the graph of:
 - (a) $y = -\sec x$
 - (b) $y = 1 - \sec x$
 - (c) $y = 1 + \sec x$
- 10 Sketch the graph of $y = \cot x$, $-\pi \leq x \leq \pi$. On the same axes, sketch the graph of:
 - (a) $y = -\cot x$
 - (b) $y = 1 - \cot x$
 - (c) $y = 1 + \cot x$
- 11 Sketch the graph of the following, stating the period of each function.
 - (a) $y = \sec \pi x$, $-1 \leq x \leq 1$
 - (b) $y = -\operatorname{cosec} 2\pi x$, $-1 \leq x \leq 1$
 - (c) $y = -\cot \frac{\pi x}{2}$, $0 \leq x \leq 2$
- 12 By adding ordinates, sketch the graphs of:
 - (a) $y = \sin \theta + \cos \theta$
 - (b) $y = 3 \sin 2\theta + 4 \sin \theta$
 - (c) $y = 2 \cos 3\theta + 3 \sin 2\theta$
 - (d) $y = \sin 2\theta - \cos \theta$
 - (e) $y = \frac{1}{2} \cos 2\theta - \sin \theta$
 - (f) $y = \sin \theta + \sin 2\theta$
- 13 By adding ordinates, sketch the graph of each function for $0 \leq x \leq 2\pi$:
 - (a) $y = \sin x + \operatorname{cosec} x$
 - (b) $y = \cos x + \sec x$
 - (c) $y = \tan x + \cot x$
 - (d) $y = \sin x + \tan x$
 - (e) $y = \cos x + \cot x$
 - (f) $y = \operatorname{cosec} x + \sec x$

EXERCISE 17.5 SOLVING INEQUALITIES GRAPHICALLY

1 Solve the following inequalities by sketching the appropriate graphs.

(a) $x + |x| \geq 2$ (b) $|x^2 - 1| > 3$ (c) $|x-1| - |x-2| < 0$ (d) $|\sin x| > \cos x, 0 < x < 2\pi$

2 Which of the following is the solution to $x + |2x| \geq 3$?

A $-3 \leq x \leq 1$ B $x \leq -3, x \geq 1$ C $x \geq 1$ D $x \leq -3$

3 Solve the following inequalities by sketching the appropriate graphs.

(a) $\frac{x-2}{x+1} < 3$ (b) $\frac{x^2+2x-6}{x-1} > 2$ (c) $x+2+\frac{4}{x-1} < -2$

(d) $\frac{x^2}{x^2-4} > -1$ (e) $\frac{1}{3x-2} < \frac{1-x}{x-6}$ (f) $\frac{1-x^2}{1+x^2} < \frac{1}{2}$

4 Graph $f(x) = x^2$ and $g(x) = |2x+1|$ on the same set of axes. Use your graph to find the values of x for which $x^2 > |2x+1|$.

5 Graph $f(x) = \frac{1}{x}$ and $g(x) = |2x-1|$ on the same set of axes. Use your graph to find the values of x for which:

(a) $\frac{1}{x} > |2x-1|$ (b) $\frac{1}{x} \leq |2x-1|$

6 Graph $f(x) = (x-1)(x+1)(x-2)$ and $g(x) = |x-1|$ on the same set of axes. Use your graph to find the values of x for which $(x-1)(x+1)(x-2) < |x-1|$.

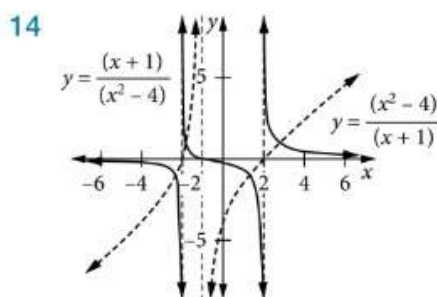
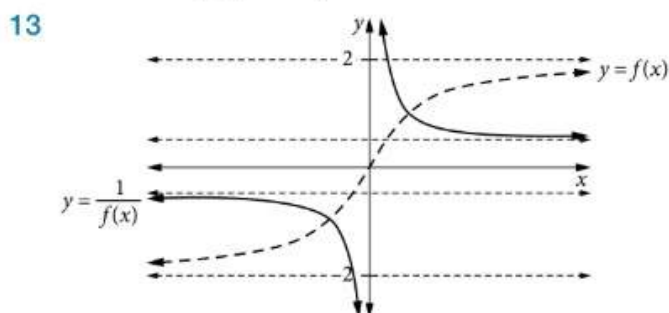
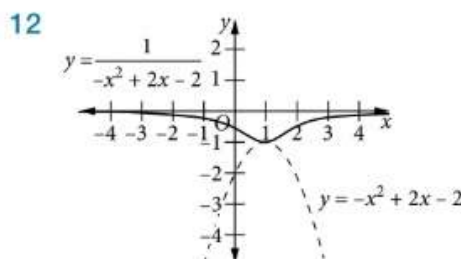
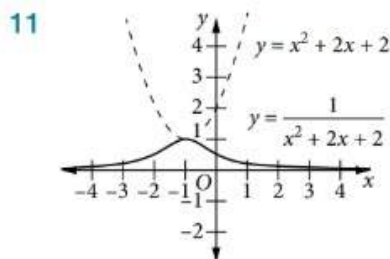
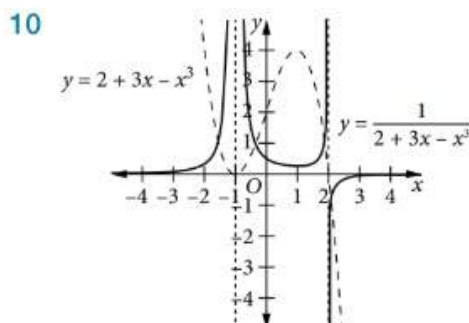
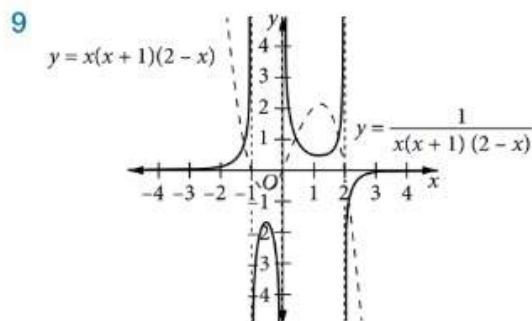
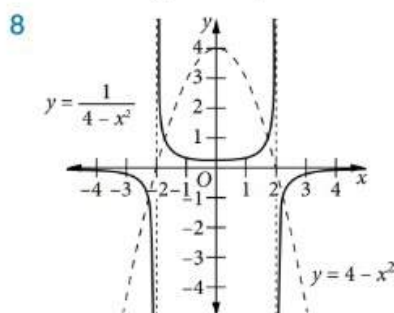
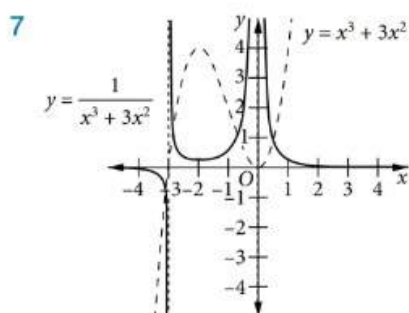
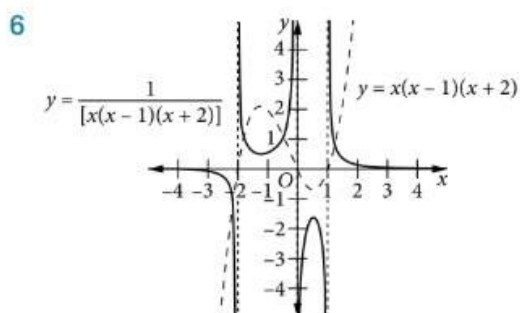
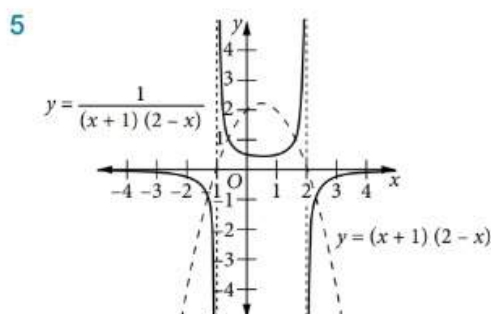
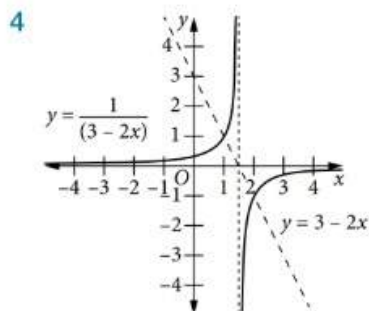
7 Graph $f(x) = 4 - x^2$ and $g(x) = x^2$ on the same set of axes. Use your graph to find the values of x for which $4 - x^2 \geq x^2$.

8 Solve the following inequalities by sketching the appropriate graphs.

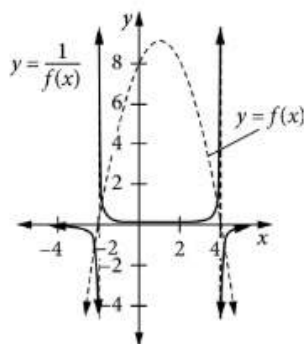
(a) $2 \sin 2x \geq 1, 0 \leq x \leq 2\pi$ (b) $\cos^2 x > 0.25, 0 \leq x \leq 2\pi$ (c) $\sin^2 x \leq 0.5, 0 \leq x \leq 2\pi$
(d) $\sin x < \cos x, 0 \leq x \leq 2\pi$ (e) $2 \sin^2 x < \tan x, 0 \leq x \leq 2\pi$ (f) $\sin x < \cos 2x, 0 \leq x \leq 2\pi$

EXERCISE 17.1

1 B 2 B 3 A

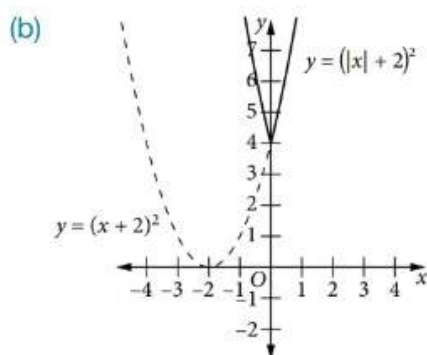
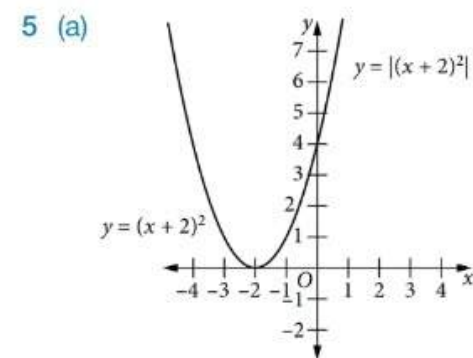
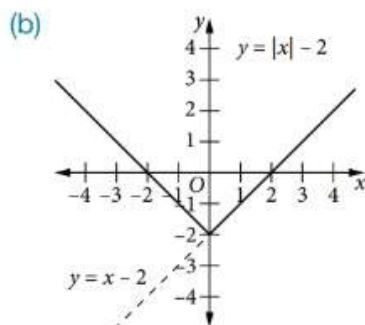
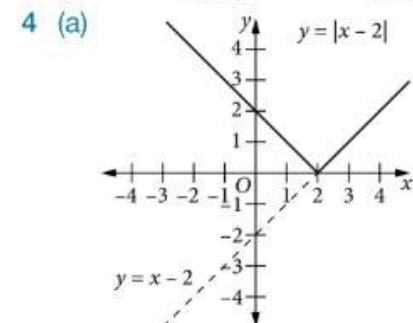


15

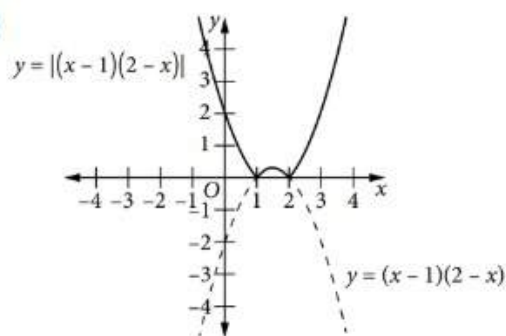


EXERCISE 17.2

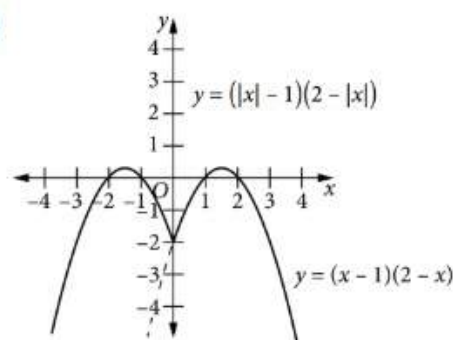
1 B 2 A 3 C



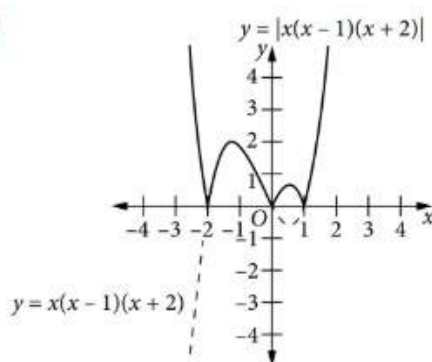
6 (a)



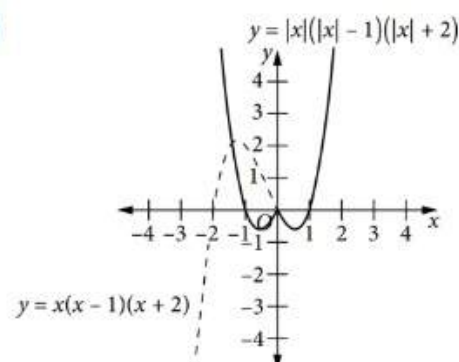
(b)



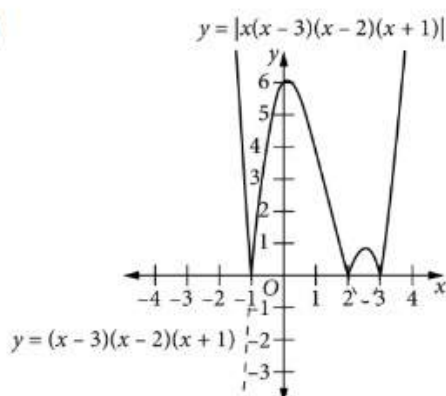
7 (a)



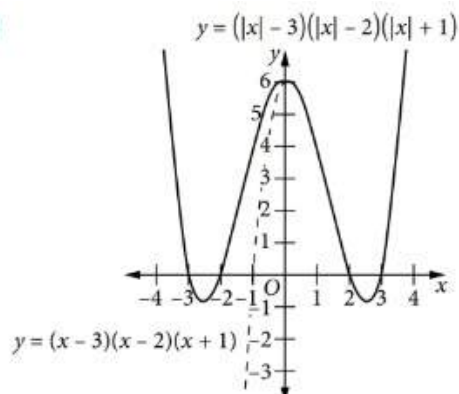
(b)



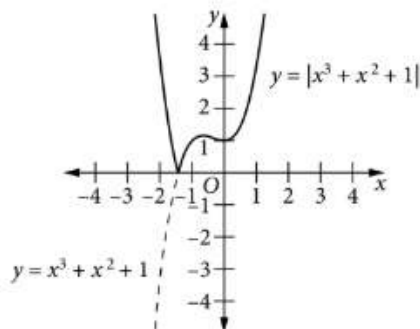
8 (a)



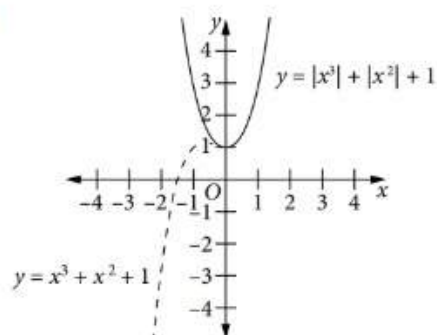
(b)



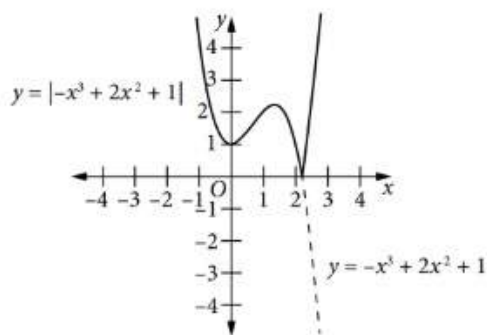
9 (a)



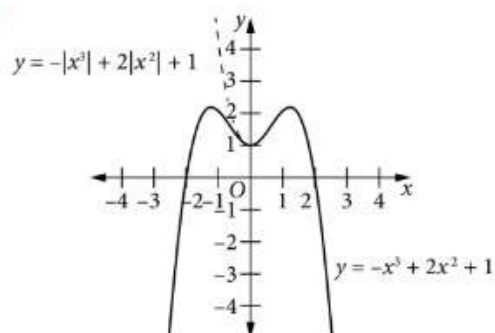
(b)



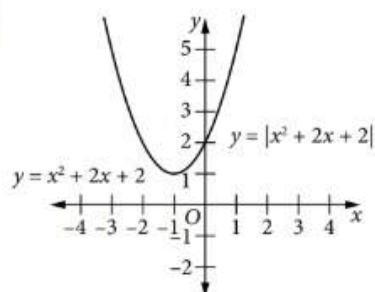
10 (a)



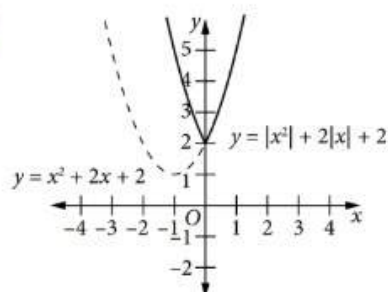
(b)



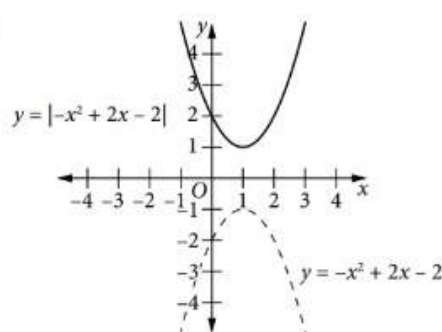
11 (a)



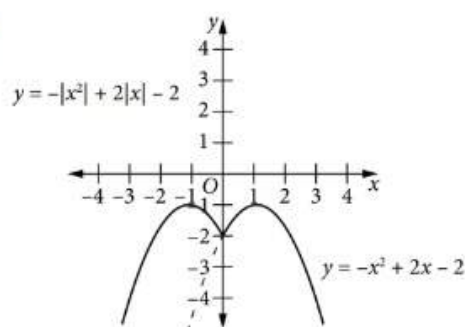
(b)



12 (a)



(b)



13 $f(x) = x^2 - c^2$

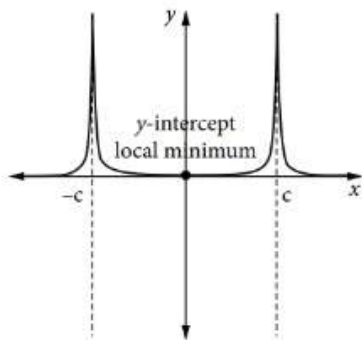
$$y = \frac{1}{|f(x)|} = \frac{1}{|x^2 - c^2|}$$

Local minimum $\left(0, \frac{1}{c^2}\right)$

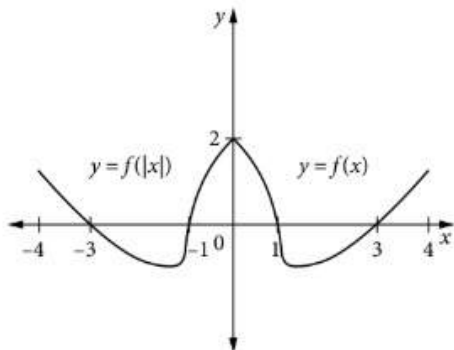
y-intercept $= \frac{1}{c^2}$

Horizontal asymptote: $y = 0$

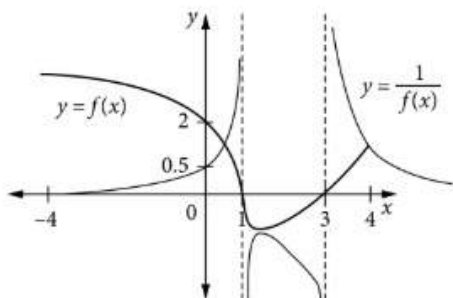
Vertical asymptotes: $x = \pm c$



14 (a) $y = f(|x|)$

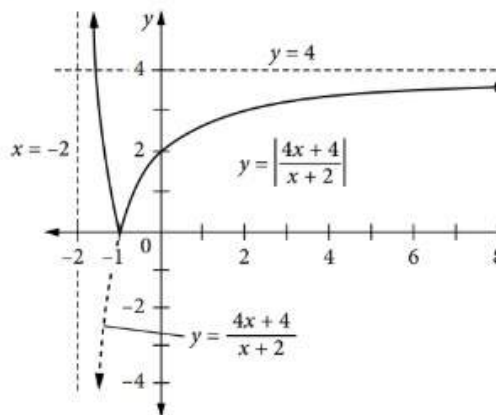


(b) $y = \frac{1}{f(x)}$

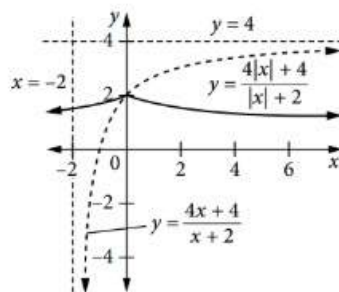


15 $f(x) = \frac{4x+4}{x+2}$, $x > -2$.

(a) $y = |f(x)|$

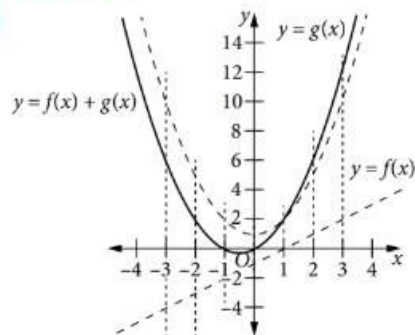


(b) $y = f(|x|)$



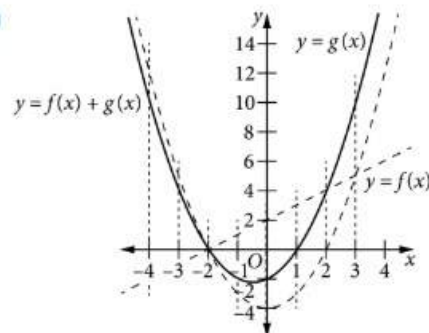
EXERCISE 17.3

1 (a)



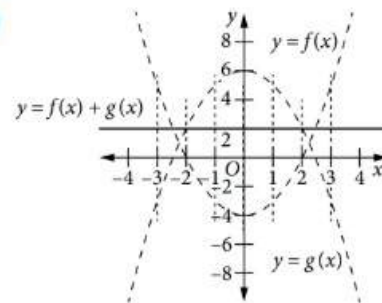
The new curve is a parabola, concave up cutting the x-axis at $(-1, 0)$ and $(0, 0)$.

(b)

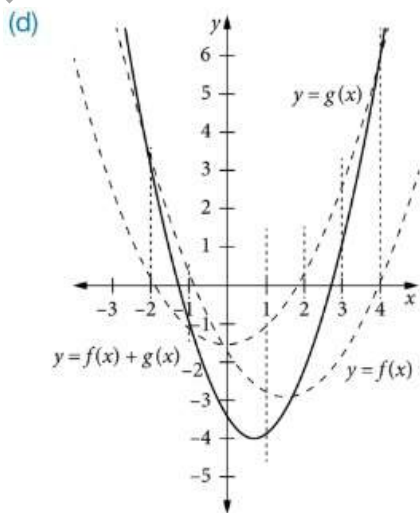


The new curve is a parabola, concave up cutting the x-axis at $(-2, 0)$ and $(1, 0)$ and the y-axis at $(0, -2)$.

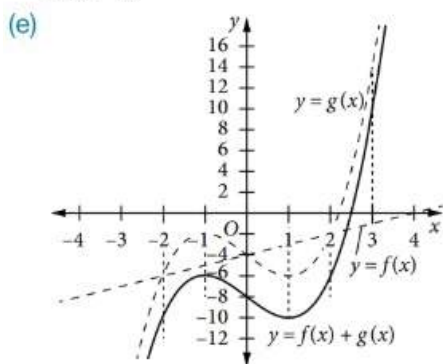
(c)



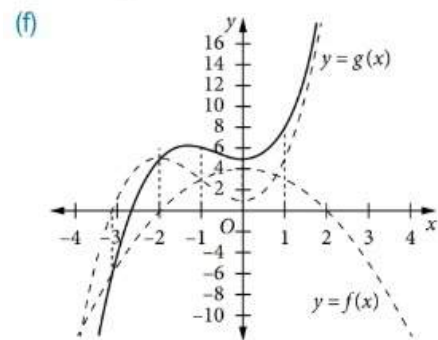
The new curve is a straight line with equation $y = 2$.



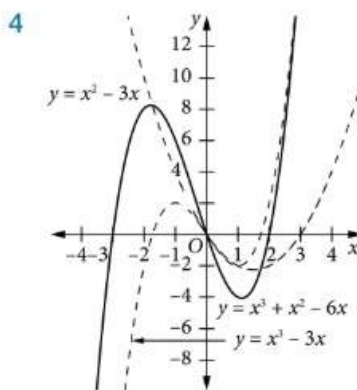
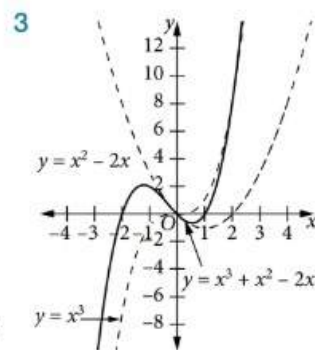
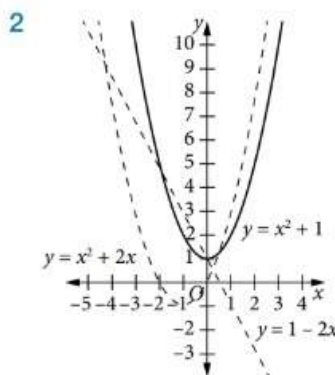
The new curve is a parabola, concave up cutting the y -axis at $(0, -4)$.



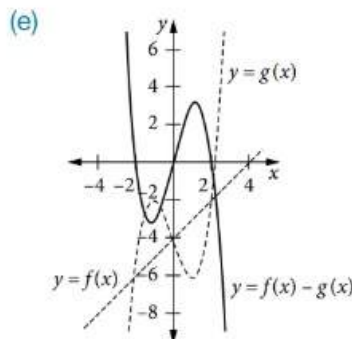
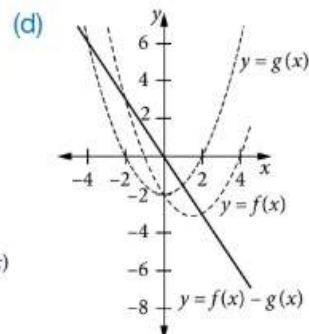
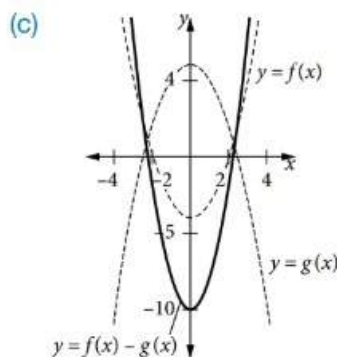
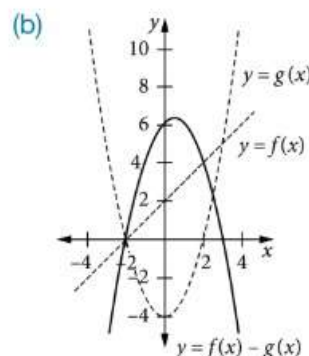
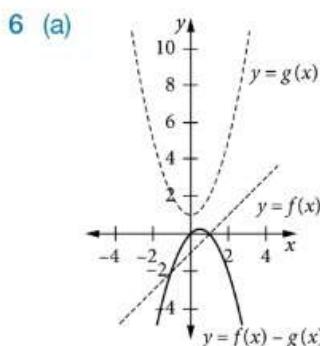
The new curve is a cubic, cutting the x -axis at about $(3.5, 0)$ and the y -axis at $(0, -8)$. It has the same shape as $y = g(x)$.



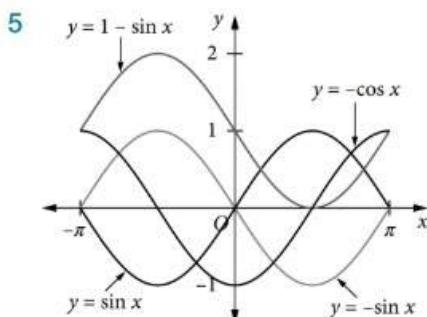
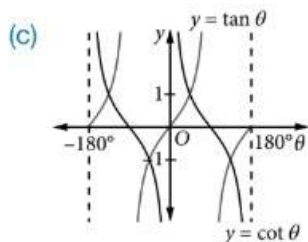
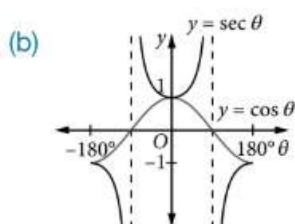
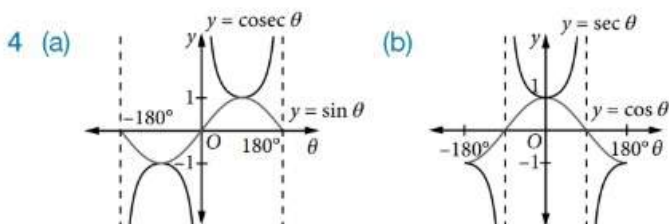
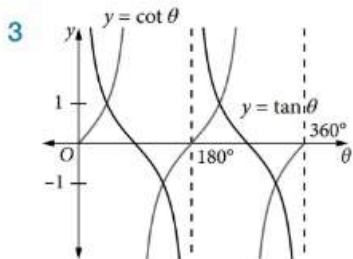
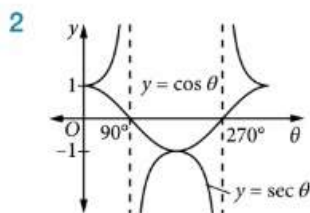
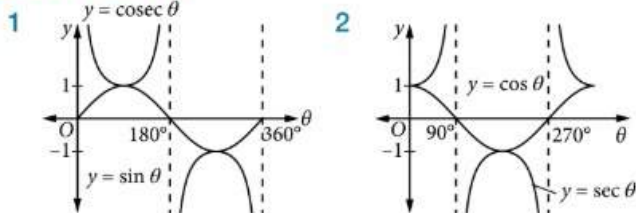
The new curve is a cubic, cutting the x -axis at about $(-2.6, 0)$ and the y -axis at $(0, 5)$. It has a similar shape to $y = g(x)$, but flattens out more between the turning points.



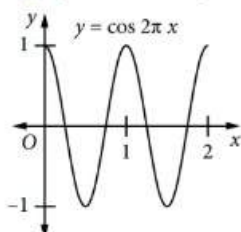
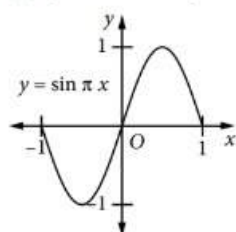
5 B



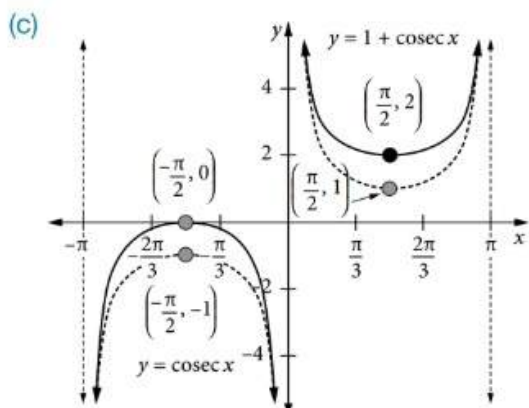
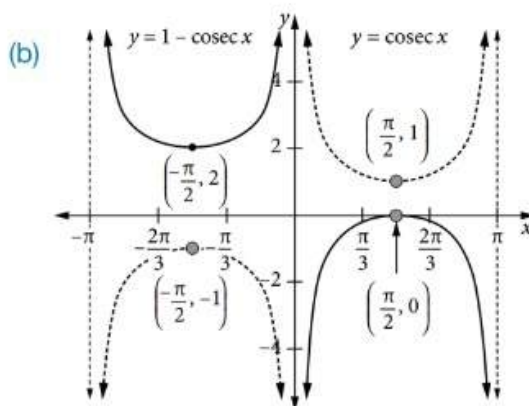
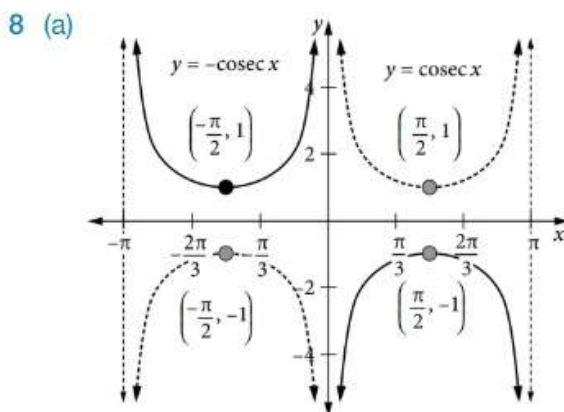
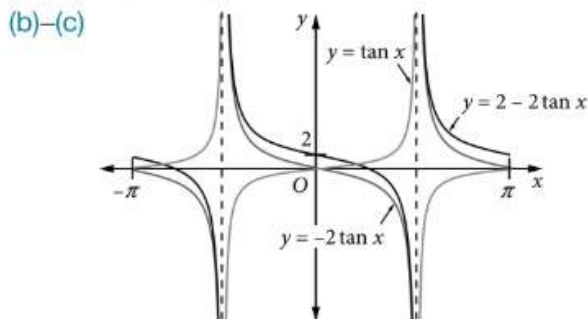
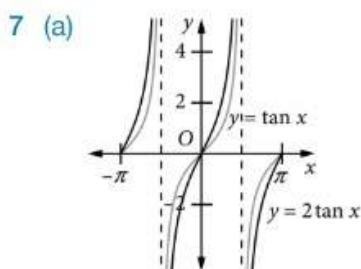
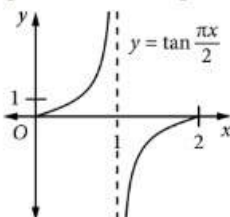
EXERCISE 17.4



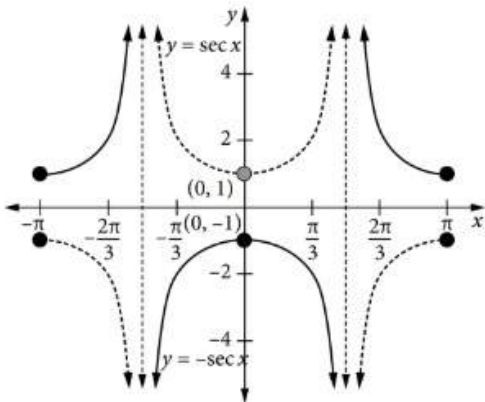
6 (a) period 2, amplitude 1 (b) period 1, amplitude 1



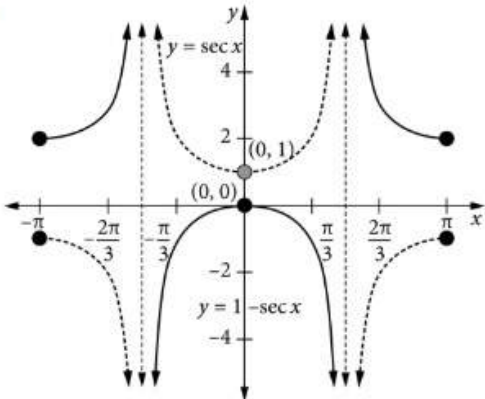
(c) period 2, no amplitude



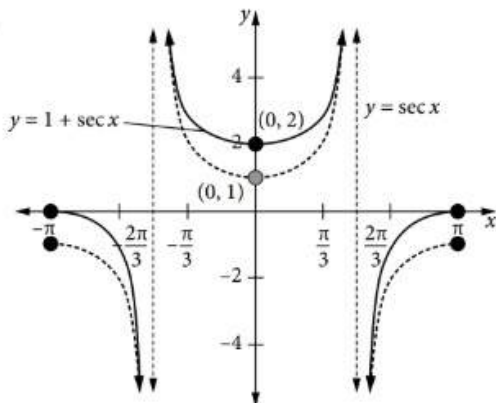
9 (a)



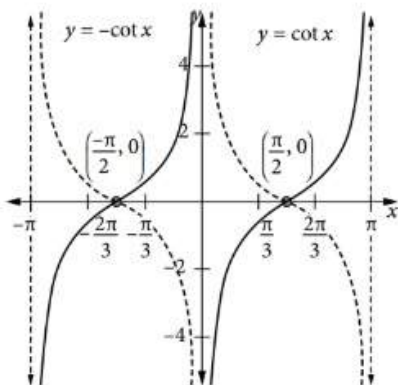
(b)



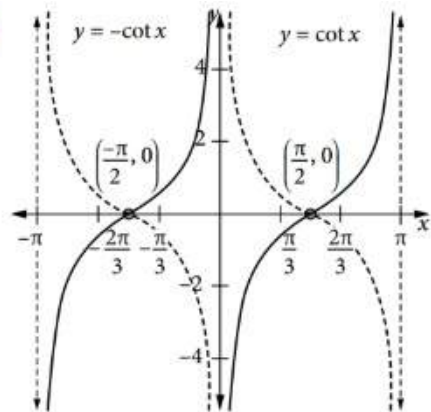
(c)



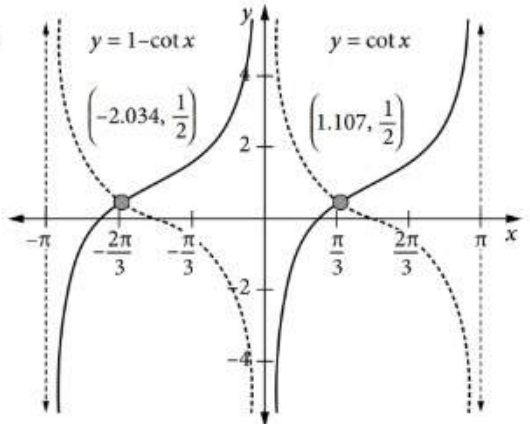
10 (a)



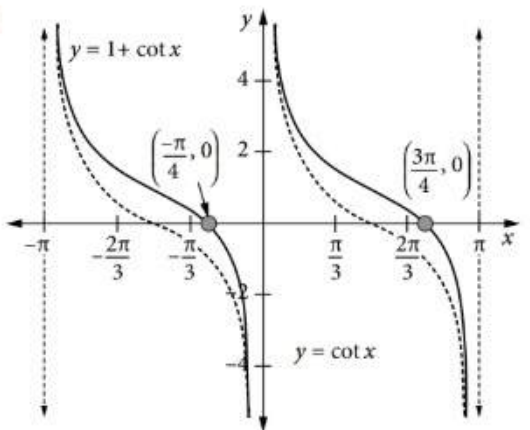
10 (a)



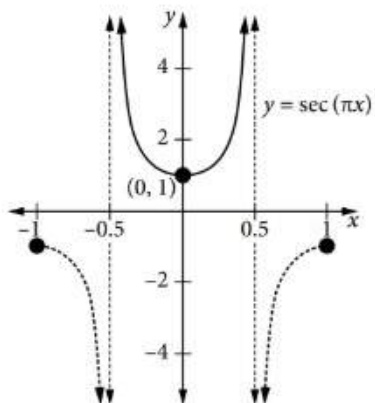
(b)



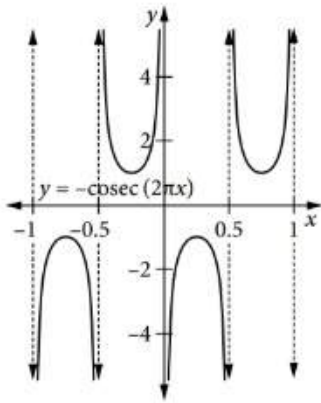
(c)



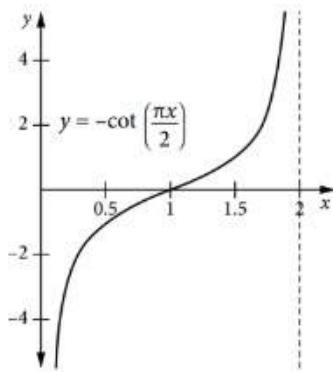
11 (a) Period: 2



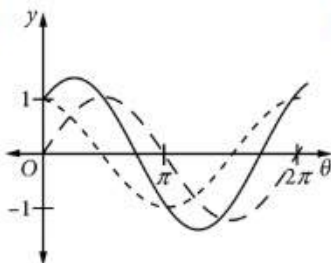
(b) Period: 1



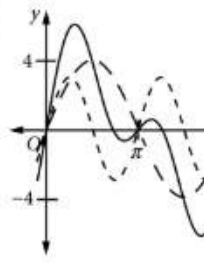
(c) Period: 2



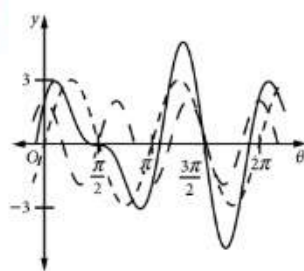
12 (a)



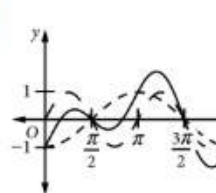
(b)



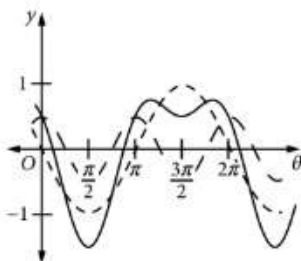
(c)



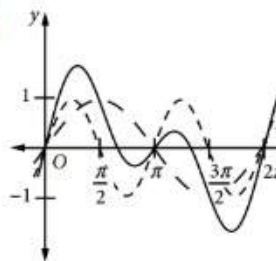
(d)



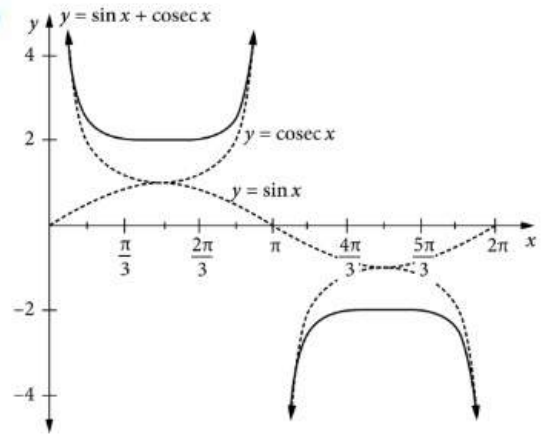
(e)



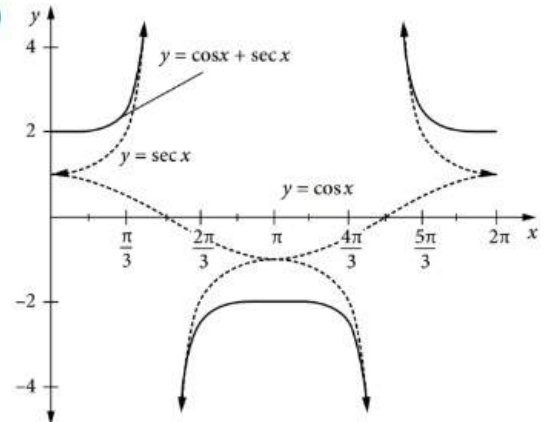
(f)



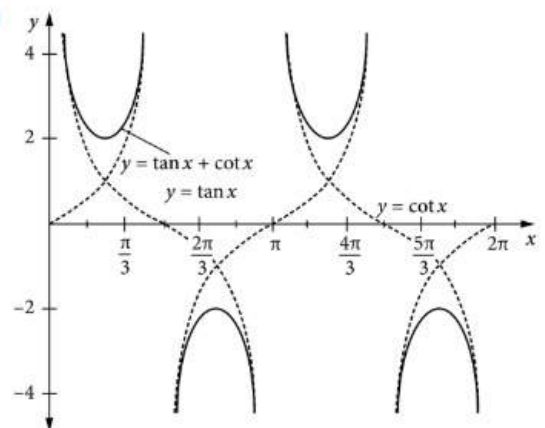
13 (a)



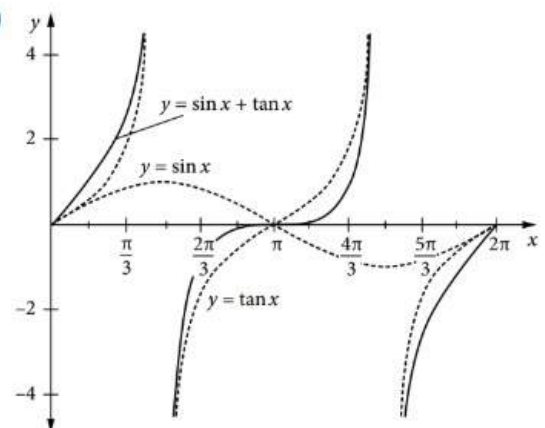
(b)

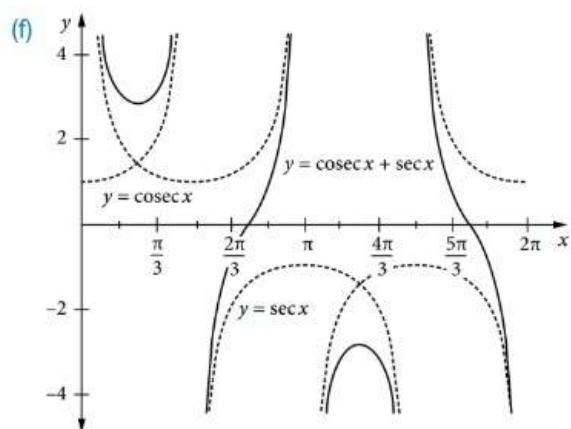
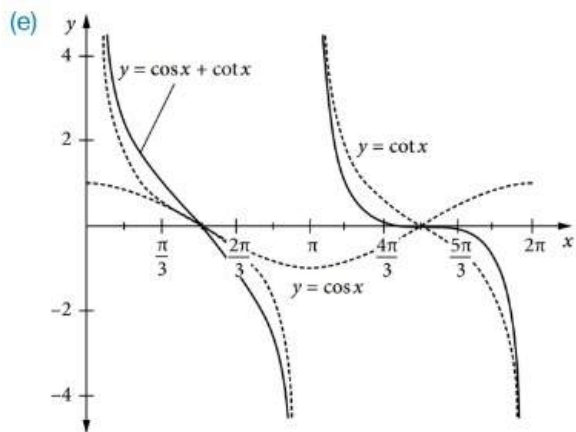


(c)



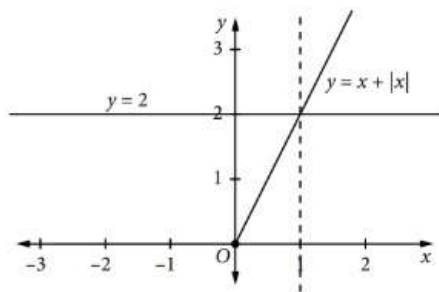
(d)



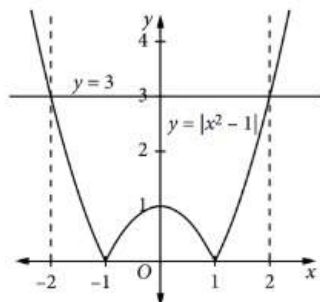


EXERCISE 17.5

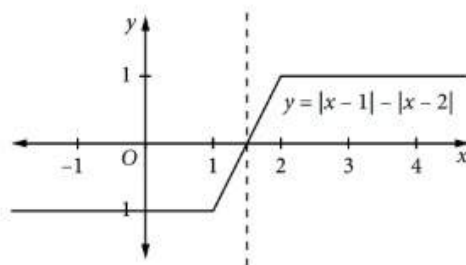
1 (a) $x \geq 1$



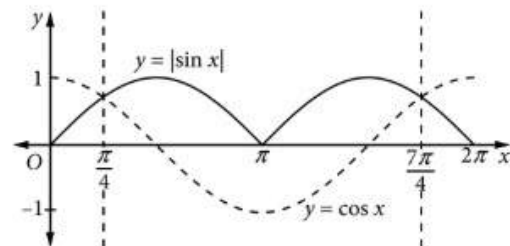
(b) $x < -2, x > 2$



(c) $x < 1.5$

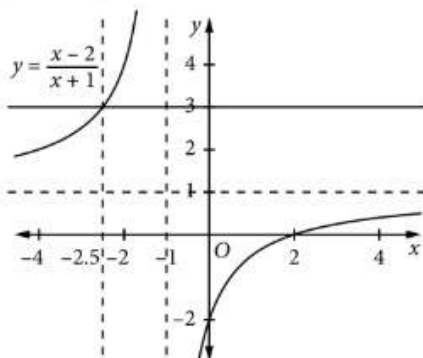


(d) $\frac{\pi}{4} < x < \frac{7\pi}{4}$

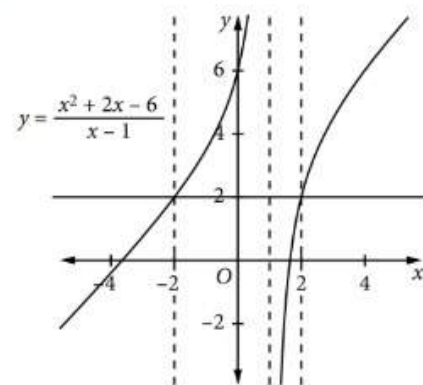


2 B

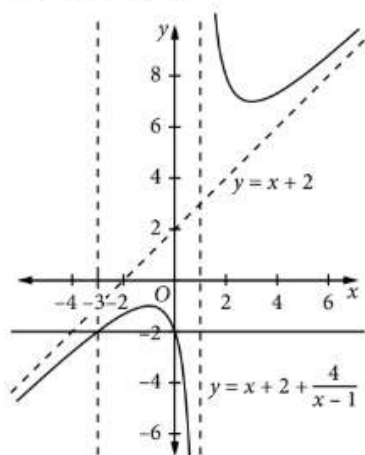
3 (a) $x < -2.5, x > -1$



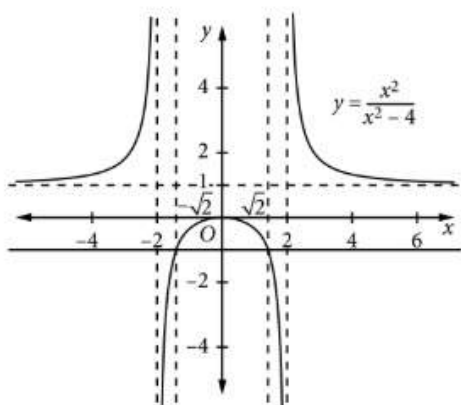
(b) $-2 < x < 1, x > 2$



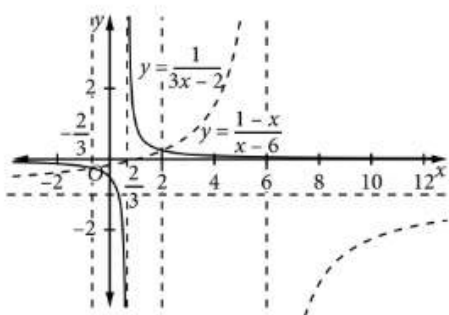
(c) $x < -3, 0 < x < 1$



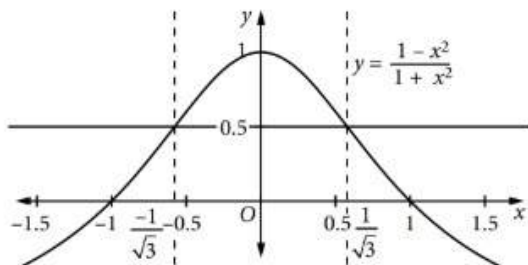
(d) $x < -2, -\sqrt{2} < x < \sqrt{2}, x > 2$



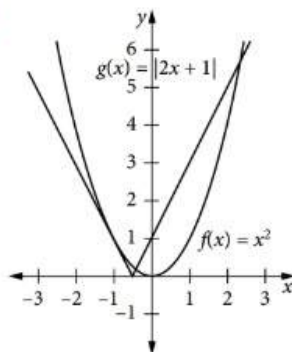
(e) $-\frac{2}{3} < x < \frac{2}{3}, 2 < x < 6$



(f) $x < -\frac{1}{\sqrt{3}}, x > \frac{1}{\sqrt{3}}$

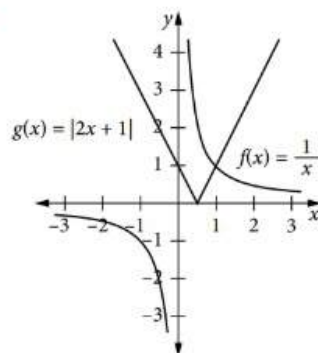


4



Solution: $0 < x < 1 - \sqrt{2}$ and $x > 1 + \sqrt{2}$

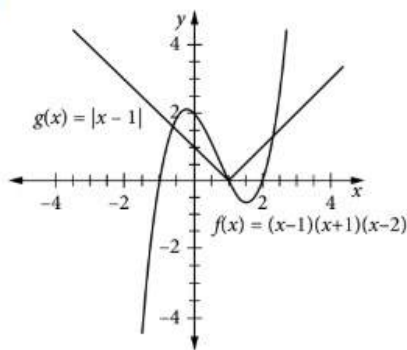
5 (a)



$\frac{1}{x} > |2x-1|$ for $0 < x < 1$

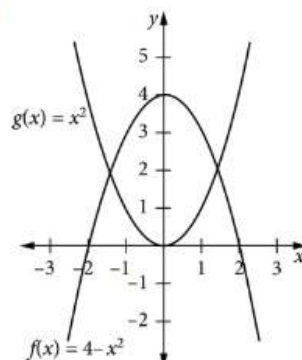
(b) $\frac{1}{x} \leq |2x-1|$ for $x < 0$ and $x \geq 1$

6



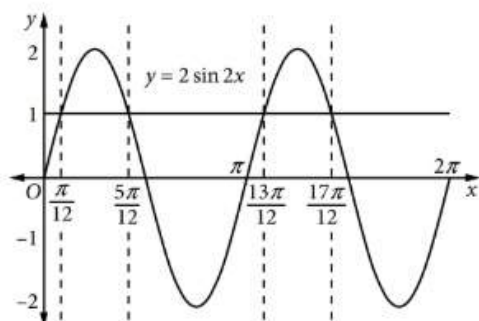
Solution is $x < \frac{1-\sqrt{5}}{2}$ and $1 < x < \frac{1+\sqrt{13}}{2}$.

7

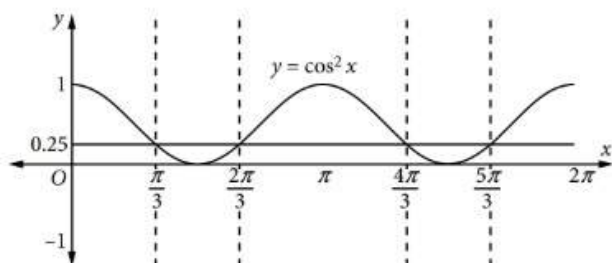


Solution $-\sqrt{2} \leq x \leq \sqrt{2}$

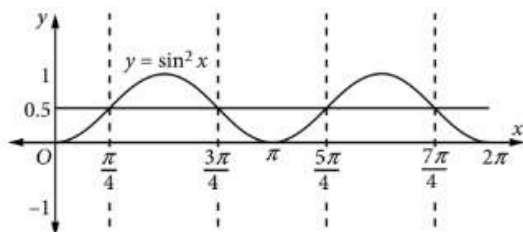
8 (a) $\frac{\pi}{12} \leq x \leq \frac{5\pi}{12}, \frac{13\pi}{12} < x < \frac{17\pi}{12}$



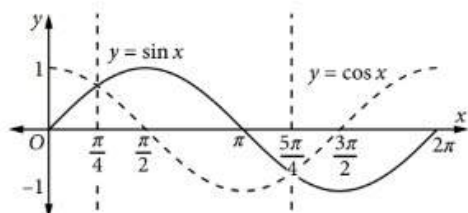
(b) $0 \leq x < \frac{\pi}{3}, \frac{2\pi}{3} < x < \frac{4\pi}{3}, \frac{5\pi}{3} < x \leq 2\pi$



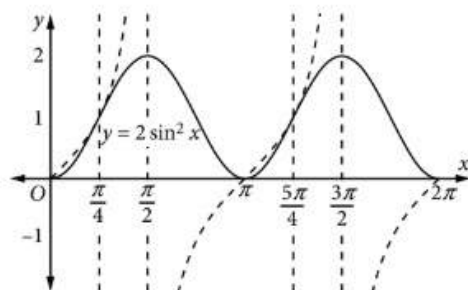
(c) $0 \leq x < \frac{\pi}{4}, \frac{3\pi}{4} < x < \frac{5\pi}{4}, \frac{7\pi}{4} < x \leq 2\pi$



(d) $0 \leq x < \frac{\pi}{4}, \frac{5\pi}{4} < x < 2\pi$



(e) $0 \leq x < \frac{\pi}{2}, x \neq \frac{\pi}{4}; \pi < x < \frac{3\pi}{2}, x \neq \frac{5\pi}{4}$



(f) $0 \leq x < \frac{\pi}{6}, \frac{5\pi}{6} < x \leq 2\pi, x \neq \frac{3\pi}{2}$

